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Uber Ride Analysis Report

Dataset Overview

This report provides a comprehensive analysis of **Uber Ride Bookings** across NCR (National Capital Region), based on the provided dataset and Jupyter Notebook developed using **PySpark**.

- **Source:** NCR Uber Ride Bookings Dataset
- **Records:** 5,000
- **Columns:** 23
- **Format:** CSV

The dataset captures detailed information about ride bookings, customer behavior, driver performance, and trip outcomes.

Key Columns:

- **Date & Time:** Date and time of ride booking
- **Booking Status:** Status of each booking (Completed, Cancelled, Incomplete)
- **Vehicle Type:** Type of vehicle selected (Mini, Sedan, SUV, etc.)
- **Pickup & Drop Location:** Starting and ending points of rides
- **Payment Method:** Payment mode used (Cash, Online, Wallet)
- **Driver Rating / Customer Rating:** Feedback metrics for quality analysis
- **Ride Distance & Booking Value:** Core indicators of ride economics.

Key Findings and Analysis

1. Ride Trends by Hour of Day

- **Insight:** Highest demand occurs during **morning (8–10 AM)** and **evening (5–8 PM)**, aligning with office commute hours.
- **Trend:** Low activity during **late night hours (12 AM–4 AM)**. This suggests commuter patterns dominate Uber ride bookings in NCR.

2. Ride Trends by Day of Week

- **Insight: Weekdays** show higher booking volumes than weekends.
- **Trend:** Slight increase on **Fridays and Saturdays**, indicating leisure travel.
  Work commute patterns strongly influence ride volume during the week.

3. Ride Trends by Month

- **Insight:** Seasonal variations observed — higher rides in **March and April**, slightly lower in monsoon months.
- **Trend:** Monthly demand fluctuations could correlate with local events or weather.

4. Booking Status Distribution

- **Insight:** Majority of bookings are **Completed**, followed by **Cancelled** by either customer or driver.
- **Trend:** Cancellations cluster around high-demand hours, indicating possible surge pricing or driver unavailability.
  Reducing cancellations could significantly improve operational efficiency.

5. Payment Method Distribution

- **Insight: Online payments** (UPI, Cards, Wallets) dominate over cash transactions.
- **Trend:** Increasing digital adoption reflects growing trust in cashless systems.

This insight supports business initiatives promoting digital incentives.

6. Vehicle Type Preferences

- **Insight: Sedan rides** are most frequent, followed by **Mini** and **SUV**.
- **Trend:** Sedans likely preferred for comfort and affordability balance.
  Vehicle supply can be optimized based on area-wise demand.

7. Ratings Analysis

- **Insight:** Both **Customer and Driver Ratings** cluster around 4–5 stars.
- **Trend:** Indicates overall satisfaction with service quality.
  This reflects strong customer engagement and driver professionalism.

Technologies Used for Analysis

The project leverages **Big Data Analytics tools** and **PySpark** for efficient data handling:

- **Python:** Core programming language
- **PySpark:** Distributed processing and large-scale data manipulation
- **Pandas:** Additional exploration and validation
- **Matplotlib:** Visualization of trends and distributions
- **Jupyter Notebook:** Interactive development and reporting

Conclusion

The **Uber Ride Analysis Project** provides actionable insights into ride-booking behavior across NCR:

- Peak demand during **morning and evening hours**
- **Weekday rides** dominate, reflecting daily commute patterns

- **Online payments** are the most preferred method
- **Sedans** lead in ride preferences
- Majority bookings are **Completed**, with manageable cancellation rates
- **High customer and driver ratings** highlight strong service experience

These findings can help Uber optimize **fleet distribution, pricing strategies, and driver allocation**.
The analysis demonstrates how **PySpark** enables scalable big data insights for ride-hailing businesses.